

Modifying the processing and handling of frozen broccoli for increased sulforaphane formation.

Frozen broccoli can provide a cheaper product, with a longer shelf life and less preparation time than fresh broccoli. We previously showed that several commercially available frozen broccoli products do not retain the ability to generate the cancer-preventative agent sulforaphane. We hypothesized that this was because the necessary hydrolyzing enzyme myrosinase was destroyed during blanching, as part of the processing that frozen broccoli undergoes. This study was carried out to determine a way to overcome loss of hydrolyzing activity. Industrial blanching usually aims to inactivate peroxidase, although lipoxygenase plays a greater role in product degradation during frozen storage of broccoli. Blanching at 86 °C or higher inactivated peroxidase, lipoxygenase, and myrosinase. Blanching at 76 °C inactivated 92% of lipoxygenase activity, whereas there was only an 18% loss in myrosinase-dependent sulforaphane formation. We considered that thawing frozen broccoli might disrupt membrane integrity, allowing myrosinase and glucoraphanin to come into contact. Thawing frozen broccoli for 9 h did not support sulforaphane formation unless an exogenous source of myrosinase was added. Thermal stability studies showed that broccoli root, as a source of myrosinase, was not more heat stable than broccoli floret. Daikon radish root supported some sulforaphane formation even when heated at 125 °C for 10 min, a time and temperature comparable to or greater than microwave cooking. Daikon radish (0.25%) added to frozen broccoli that was then allowed to thaw supported sulforaphane formation without any visual alteration to that of untreated broccoli. © 2013 Institute of Food Technologists®